PS-049 — Week 2 Design Summary

Returns and Reverse Logistics Grading Engine

Architecture

models.py defines the domain entities and enumerations. validation.py parses and validates incoming order data and timestamps. eligibility.py makes the return decision. state_machine.py is the single source of truth for allowed state changes. engine.py orchestrates the case lifecycle. grading.py computes condition scores, grade, route and recovery. refunds.py calculates deductions, settlement and loss allocation. reports.py builds case and weekly reports. cli.py reads a JSON command file and writes a JSON report.

Main happy path

RETURN creates a REQUESTED case. Eligible cases become APPROVED then PICKUP_SCHEDULED. A successful PICKUP moves to PICKED. RECEIVE verifies the serial, records RECEIVED, computes score and grade, then routes the unit. Grade A becomes RESTOCKED; grade B becomes REFURB_IN_PROGRESS and then RESOLD or LIQUIDATED according to QC; grade C becomes LIQUIDATED; grade D becomes SCRAPPED. Refund and loss records are calculated independently of product recovery.

Special paths

• Ineligible return: REJECTED.

• Serial mismatch: FRAUD_REVIEW and no refund.

• Two failed pickup attempts: CUSTOMER_UNAVAILABLE; no refund. One reopen is allowed within three days.

• Grade C/D: refund waits for manager approval.

• Exchange: replacement is marked shipped on successful pickup; no cash refund; C/D creates wallet adjustment.

State safety

transition() checks the current state against ALLOWED. If a transition is not listed, it raises ValueError without changing the case. Terminal states have no outgoing ordinary transitions. Reopening is handled by a dedicated method that checks the one-time limit and time window.

Validation strategy

The input layer rejects missing order fields, malformed or timezone-naive timestamps, negative amounts, invalid payment types and unknown enums. The engine rejects unknown order/case IDs, duplicate case IDs, invalid pickup outcomes, excess attempts, non-consecutive attempts, invalid reopen requests and illegal transitions. The test suite covers key boundaries and business scenarios.